



DEEP LEARNING · MEDICAL IMAGING · EXPLAINABLE AI

Early-Stage Alzheimer's Detection

Using Deep Learning on Brain MRI

Binary classification of Non-Demented vs. Very Mild Demented MRI scans with a custom CNN, Grad-CAM explainability, and a deployed Streamlit application.

Capstone Project Presentation- by Muhsina v s

Why Early-Stage Detection Is Hard

Alzheimer's is progressive. The clinically hardest boundary sits between a healthy scan and the earliest signs of decline — which is exactly the boundary this project targets.



Class 0 — Non-Demented

No measurable cognitive decline visible in the MRI scan.



Class 1 — Very Mild Demented

Subtle, early-stage cognitive decline — the hardest signal to catch.



Design Priority: Minimize False Negatives

Missing a Very Mild Demented case is far more costly than a false alarm. Modeling and evaluation were built to be sensitivity-focused — recall on the disease class is treated as the primary success metric, not accuracy alone.

Kaggle Brain MRI Dataset

Alzheimer's Disease Multiclass Images Dataset (Equal and Augmented) — downloaded via the Kaggle API and filtered down to the two classes this project targets.

24,000

Total MRI Images Used

12,800

Non-Demented Images

11,200

Very Mild Demented Images

Dataset Properties



Sourced from Kaggle via the Kaggle API, downloaded programmatically.



Only the two required classes were extracted from the archive — reduces disk & RAM footprint.



Images are skull-stripped (non-brain tissue removed) MRI slices.



Dataset is pre-balanced and augmented across classes.

From Raw Archive to Model-Ready Data



Step 1

Download & Extract

Pulled the dataset via Kaggle API; selectively extracted only NonDemented / VeryMildDemented folders, then deleted the zip to save disk space.



Step 2

Build DataFrame

Indexed every image into a pandas DataFrame with filepath, filename, and class label — 24,000 rows total.



Step 3

Patient-Level Split

Grouped images by patient ID before splitting, so no patient's scans leak across train / val / test.



Step 4

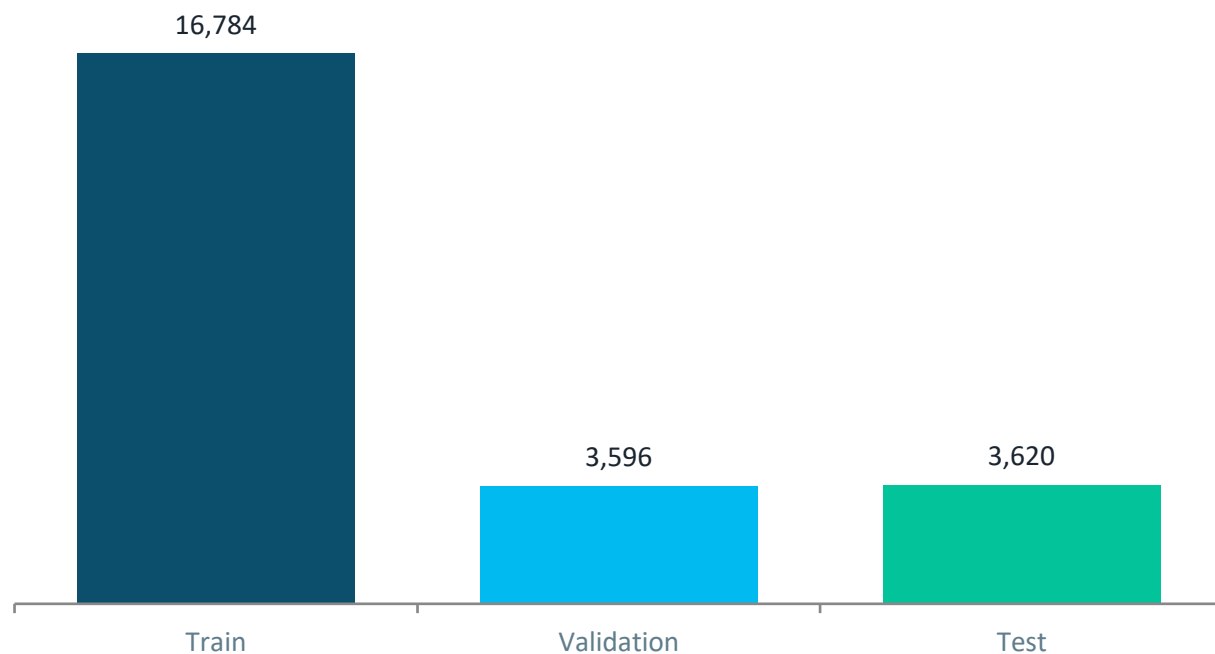
Preprocess

Resized every image to 224×224 and normalized pixel values to the [0, 1] range.

Patient-Level Train / Val / Test Split

Splitting was done on unique patient IDs (70% / 15% / 15%) rather than on raw images, preventing data leakage from the same patient appearing in multiple splits.

Resulting Image Counts



16,499

Train Patients

3,535

Validation Patients

3,536

Test Patients

Efficient Loading with tf.data

Images are streamed from disk rather than held fully in memory, using TensorFlow's tf.data API for a scalable, GPU-friendly pipeline.



Why this matters

Streaming + prefetching kept memory usage low and avoided the runtime crashes seen when loading all 24,000 images into RAM at once.

Baseline CNN Architecture

A compact convolutional network establishes a performance floor and a fast, deployable baseline before any transfer-learning comparison.



Model Facts

Total Params

109,889

Trainable

109,889

Output

Sigmoid (binary)

Loss

Binary Cross-Entropy

Framework

TensorFlow / Keras

Training Configuration



Optimizer

Adam, learning rate = $1e-4$



Loss Function

Binary Cross-Entropy



Metric Tracked

Accuracy (val_loss for callbacks)



Early Stopping

monitor = val_loss, patience = 5, restores best weights



LR Scheduling

ReduceLROnPlateau — factor 0.5, patience 3



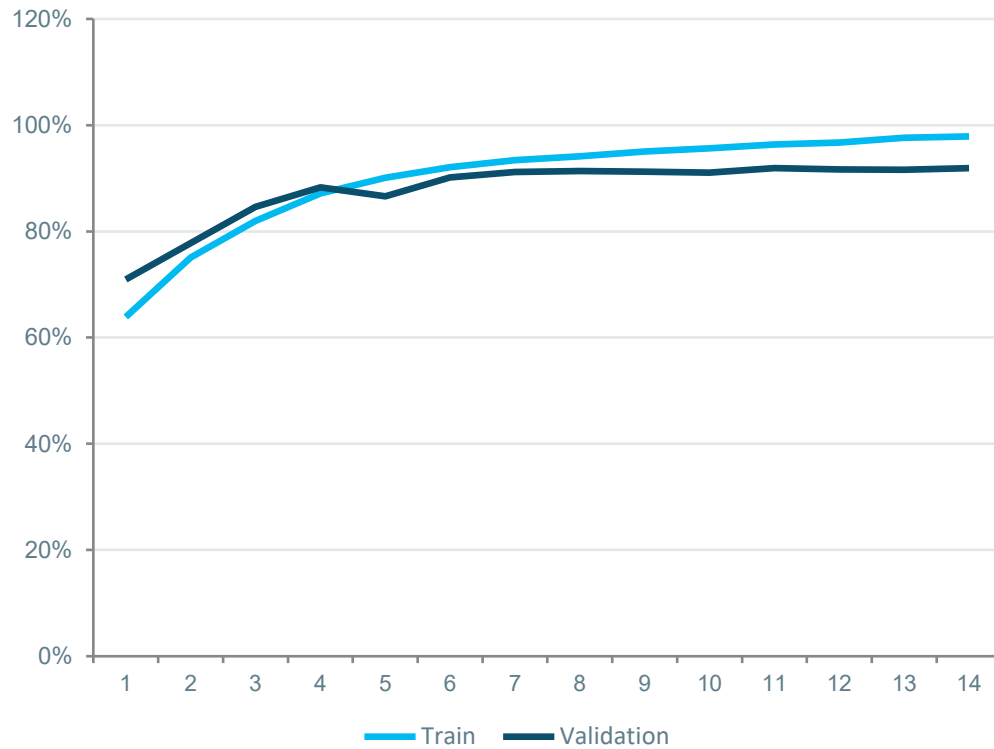
Epochs / Batch

Up to 30 epochs, batch size 16

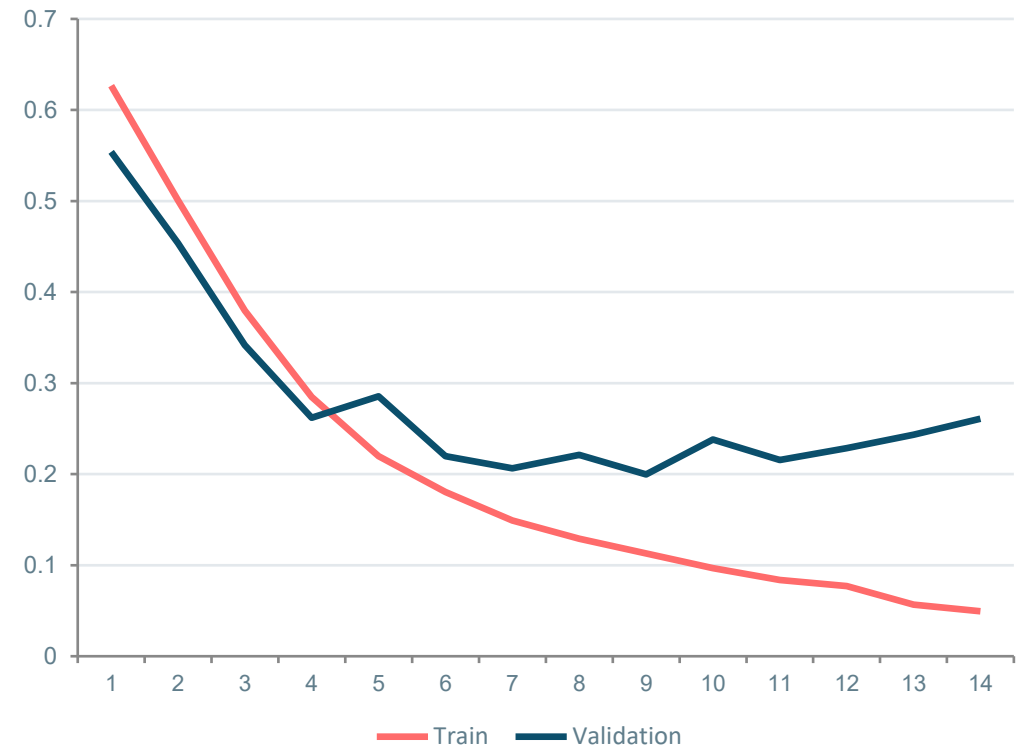
Training Curves

Accuracy and loss logged across epochs 1–14 of the callback-monitored training run.

Accuracy



Loss



Test Set Performance

Accuracy alone is misleading for a medical task — precision, recall, F1, the confusion matrix, and ROC-AUC were all reviewed, with recall on Very Mild Demented treated as the priority metric.

92.54%

Accuracy

91.84%

Precision

91.84%

Recall (Sensitivity)

91.84%

F1-Score

Confusion Matrix

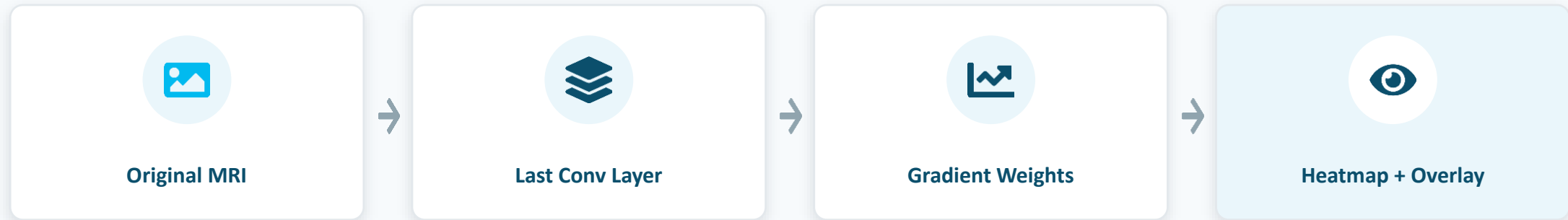
Reviewed on the held-out test set to separate false negatives (missed Very Mild Demented cases) from false positives — the clinically costlier error type.

ROC Curve & AUC

Plotted true-positive vs. false-positive rate across thresholds to assess overall separability between the two classes, independent of a single cut-off.

Seeing What the Model Sees — Grad-CAM

Deep models are often black boxes. Grad-CAM (Gradient-weighted Class Activation Mapping) visualizes which regions of each MRI slice drove the prediction.



Clinically Meaningful Attention



Hippocampal region

Strongly associated with early hippocampal atrophy.



Medial temporal lobe

One of the first regions to show structural change.

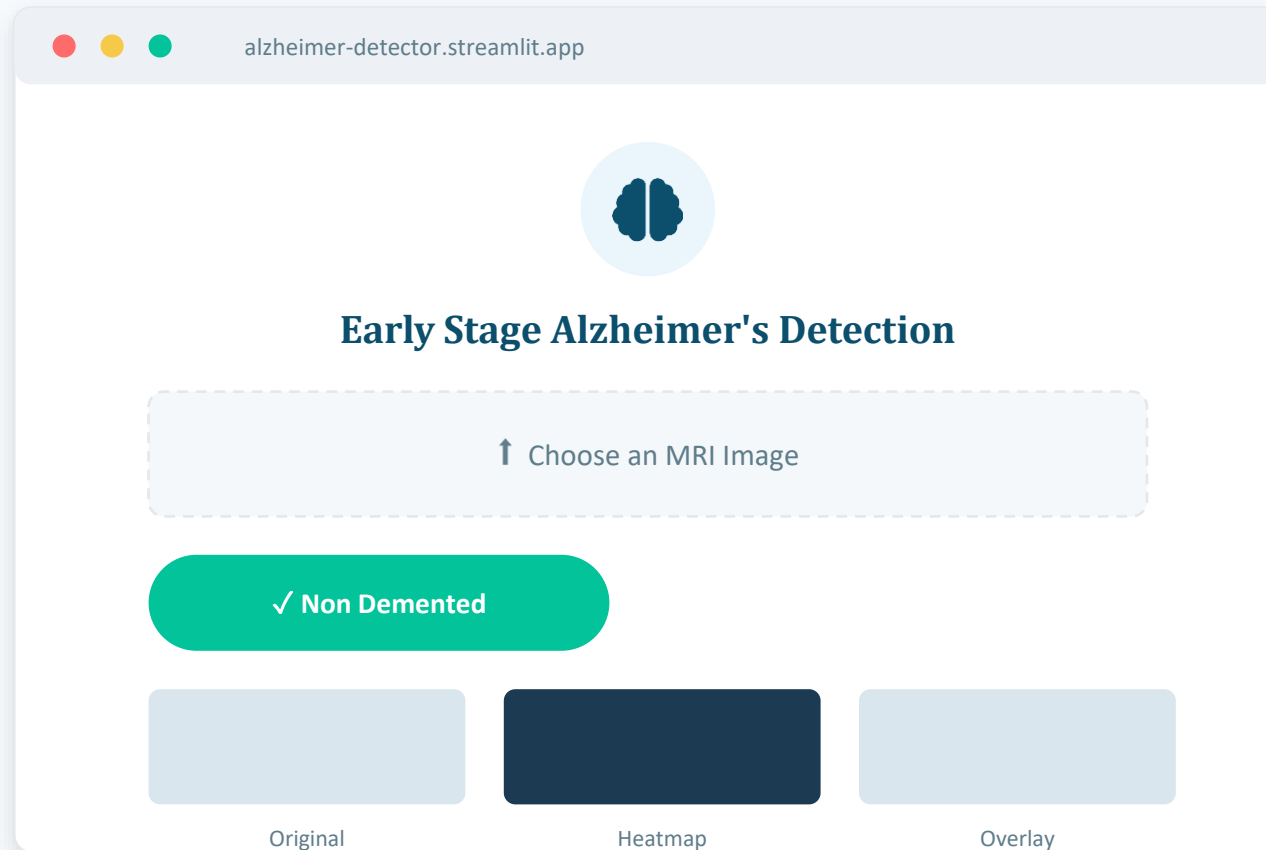


Ventricular enlargement

Often visible early in cognitive decline.

From Notebook to Streamlit Web App

The trained model (`alzheimer_model.keras`) was wrapped in a Streamlit application so predictions and Grad-CAM explanations are available through a simple browser UI — no notebook required. [Early Stage Alzheimer's Detection · Streamlit](#)



Upload MRI Image

Drag-and-drop JPG/PNG upload straight from the browser.



Prediction + Confidence

Class label, sigmoid confidence score, and probability shown instantly.



Grad-CAM Side-by-Side

Original, heatmap, and overlay images shown together for transparency.



Cloud-Hosted

Deployed on Streamlit Community Cloud, accessible from any browser.

Key Takeaways & Future Work



What Worked

- Patient-level splitting avoided data leakage across train/val/test.
- A lightweight 3-block CNN reached 92.5% test accuracy with balanced precision & recall.
- Grad-CAM confirmed the model can be inspected, not just trusted blindly.
- The full pipeline was packaged into a working Streamlit app for real interactive use.



Limitations & Next Steps

- Currently binary only — extend to the types of dementia.
- Grad-CAM heatmaps are coarse (28×28 upsampled) given the shallow conv stack — deeper/pretrained backbones would sharpen localization.
- Compare against transfer-learning baselines (e.g. ResNet50) mentioned as a follow-up in the original workflow.
- Validate on an external, out-of-distribution MRI dataset before any clinical framing.



Thank You

Questions & Discussion

Early-Stage Alzheimer's Detection Using **Deep Learning on MRI**
By muhsina vs